

Classifying Alzheimer's Disease Using Convolutional Neural Networks on Multi-Channel EEG Data

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Abstract—Alzheimer's disease is a long-term illness characterized by a decline in abilities and the development of dementia. The main objective of this study is to explore the potential of using EEG-based classification for identifying and comprehending Alzheimer's disease in its stages. EEG, which measures brain activity, offers insights into the functioning of neurons. This survey presents a careful examination of existing modalities, feature extraction techniques, datasets, machine learning methods, and validation approaches used in Alzheimer's disease detection. For our research, we used a dataset that included EEG recordings from individuals diagnosed with Alzheimer's disease, mild cognitive impairment, and healthy individuals for comparison. EEG recordings from 19 channels were utilised as input features to investigate classification performance using a fine-tuned pretrained CEEDRestNet model. Additionally, 10 channels were extracted as features to be classified by a customized 1D ResNet-18 convolutional neural network. To evaluate the effectiveness of these approaches, we assessed accuracy, sensitivity, specificity and precision metrics. The experimental framework was designed to test the capability of differentiating between Alzheimer's patients and healthy individuals through neural signal analysis.

Index Terms—Deep Learning, Alzheimer's Disease, EEG, Mild Cognitive Impairment, Electroencephalography

I. INTRODUCTION

Alzheimer's disease is a long-term condition that impacts the population across the globe, contributing to 60–80% of dementia cases that are diagnosed [2,3]. It is characterized by the accumulation of tau tangles and beta-amyloid plaques in the brain which over time contribute to the loss of neurons and synapses, causing a deterioration in capacities [4,5]. Typically, Alzheimer's disease begins with disorientation and memory loss, which deteriorates into serious cognitive problems, including language, cognitive, and executive function difficulties. Sadly, there is currently no treatment for

Alzheimer's. The existing treatments aim to alleviate symptoms and slow the progression of the illness.

Fig. 1 visually demonstrates how the number of deaths caused by Alzheimer's disease has changed from 1999 to 2021. The data [6] offers information on these deaths over this time. The figure clearly shows a trend in the number of deaths, with some fluctuations occurring over the years. Specifically, there was an increase in deaths, from 44,536 in 1999 to 134,242 in 2021. These statistics highlight the impact that Alzheimer's disease has had on the population over the years. Addressing this growing burden necessitates efforts to understand, prevent, and treat Alzheimer's disease for the benefit of affected individuals and their families.

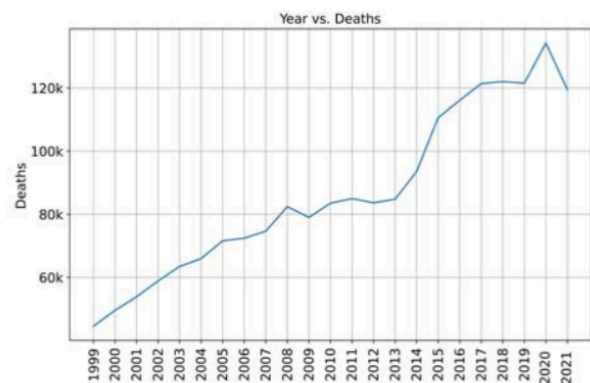


Fig. 1. Total deaths by Alzheimer's disease.

The electroencephalogram (EEG) records the voltage variations resulting from the collective firing of neurons to quantify the electrical activity of the brain. It offers a clear, cheaper and non-invasive view into the neuronal dynamics

and functioning of the brain. Since EEG can record the temporal dynamics and synchronisation of brain activity, which are impacted by the degenerative processes of AD, it provides special advantages in this context. Research has uncovered distinctive EEG rhythms and irregularities seen in AD patients. These include changes in connection metrics like coherence and phase synchronisation between various brain regions, as well as a decrease in alpha and theta band power and an increase in delta and beta band power [7]. In AD, connectivity between different brain regions is disrupted, and underlying neuronal dysfunction is reflected in these anomalies in the EEG [8].

Modern computational methods and machine learning algorithms have improved the study of EEG in AD even more. With the aid of these techniques, it is possible to identify relevant characteristics in EEG signals and create prediction models for the identification of AD. When it comes to utilising EEG data to distinguish AD patients from healthy people and recognise patterns associated with the disease, machine learning techniques have demonstrated potential.

The research aims to leverage the capabilities of EEG and machine learning to enhance the early detection and comprehension of AD.

II. METHODOLOGY

The process of diagnosing Alzheimer’s disease using EEG includes steps as shown in **Figure 2**. First, the data is converted to CSV, then preprocessed by normalising and filtering it. Then, meaningful features are selected using mutual information correlation and then extracted. Finally, the custom model and the fine-tuned model are used for classification. To evaluate the performance of the classifier, metrics such as accuracy, precision, recall, and F1 score are considered. By following this approach, EEG signals can be effectively transformed into features that enable the diagnosis and classification of Alzheimer’s disease.

A. Dataset

The CAUEEG dataset [9] used contains 1379 EEG recording signals acquired from 1155 patients at Chung-Ang University Hospital from 24th August 2012 to 12th March 2020. EEG. The other two channels are electrocardiogram (EKG or ECG) and photic stimulation. The EEG recording was conducted according to the International 10-20 system (electrode locations: Fp1, F3, C3, P3, O1, Fp2, F4, C4, P4, O2, F7, T3, T5, F8, T4, T6, FZ, CZ, and PZ) with linked earlobe referencing. The EEG signals of subjects were measured while they were lying comfortably in bed but awake, and a qualified technician in the hospital supervised the recording process. The signals

were then converted to the common average referencing and saved in European data format (EDF) on disks.

For each EEG recording, the CAUEEG dataset provides i) the event history that occurred while recording, ii) the patient’s age at the time, and iii) the corresponding diagnosis label decided by neurologists based on neuropsychological examination and criteria. Fig. 3 visualises the occurrence and types of diagnostic labels in the dataset.

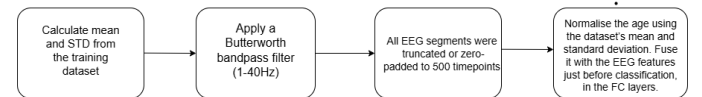
B. Pre-Processing

To make the EEG data convenient to use, we have converted it to CSV files. Data from subjects with conditions other than AD were filtered out, and files were renamed with the class label as the first digit. For the custom model, 9 channels were ruled out, and each of the CSV files were brought down to a reduced dataset of 10 channels.

1) *For the fine-tuned pretrained RESNET-18 model:* All EEG recordings were first visually reviewed to ensure artefact-free segments were selected for analysis. Each CSV file contained 22 channels comprising 19 EEG channels and 3 additional physiological or auxiliary signals, recorded at a sampling frequency of 250 Hz. For each participant, 500-sample epochs (~2 seconds) were extracted and preprocessed as follows:

To reduce low-frequency drift and high-frequency noise, a 5th-order zero-phase Butterworth bandpass filter was applied to all channels with cutoff frequencies set between 1.0 Hz and 40.0 Hz. Each channel was filtered independently to preserve temporal dynamics while attenuating physiological and environmental artefacts. Following filtering, the signal length was standardised to 500 time points across all samples by either truncation or zero-padding. For robust model performance, all signals were normalised using dataset-specific statistics. A random subset of 100 training samples was selected to compute the global mean and standard deviation across all EEG channels.

Specifically, the first 20 channels—corresponding to EEG activity—were normalised to zero mean and unit variance:



$$[some\ weird\ ahh\ equation] \quad (1)$$

where μ [wondering how to incorporate MEW] denote the

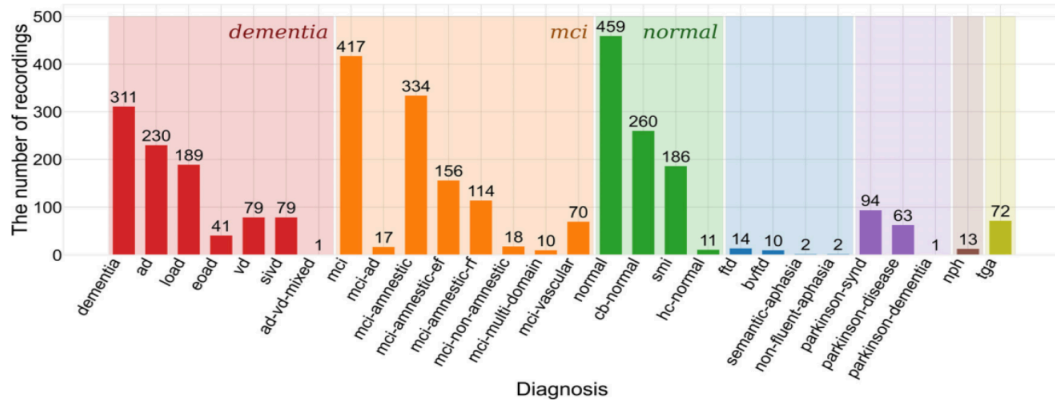


Fig. 2. Bar chart showing the number of EEGs with each symptom. The numbers above each bar denote the occurrence. The color boxes represent the groups of diagnoses. The CAUEEG dataset has no intergroup co-occurrence except for *parkinson-dementia*, which is simultaneously included in the *dementia* and *parkinson-synd* groups. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mean and standard deviation of EEG data, and $\epsilon=10^{-8}$ is a small constant to ensure numerical stability.

[another equation] (2)

where μ [mew again, figure this out] represents the mean and standard deviation of age across training samples.

The preprocessed signals, paired with their corresponding normalised age values and diagnostic labels (Healthy Control, Mild Cognitive Impairment, Alzheimer's Disease), were subsequently used for model training and evaluation. In addition to the EEG input, patient age was incorporated as an auxiliary input to the classification model. Age values were standardised using the training subset's population statistics:

2) *For the custom model:* To ensure consistency across recordings and facilitate model convergence, all EEG signals were subjected to a standardised preprocessing pipeline. From each raw EEG file, the first 10 channels were selected and transposed into a [channels $\times$ time] format. To handle variable-length recordings, signals exceeding 2000 time points were randomly cropped, while shorter ones were zero-padded to maintain a fixed input size. Finally, z-score normalisation was applied channel-wise to remove baseline shifts and scale differences. This preprocessing helped reduce noise and ensured that the model learned robust temporal patterns across all subjects.

Feature extraction of 10 channels was done by testing the data with Pearson, Spearman, Point Biserial, Mutual information and ANOVA F correlation functions, of which mutual information was deemed best fit for our data. The mutual information correlation calculates:

$$MI(X, Y) = \sum_{x \in X} \sum_{y \in Y} P(x, y) \cdot \log \left(\frac{P(x, y)}{P(x) \cdot P(y)} \right) \quad (3)$$

where $P(X, Y)$ is the probability of EEG value x and label Y occurring together, and $P(X)$ and $P(Y)$ are their independent probabilities.

	Channel	Pearson	Spearman	PointBiserial(AD vs others)	Mutual Info	ANOVA F	Average
0	FZ-AVG	0.009921	0.086611	0.050027	0.502391	0.774636	0.284717
1	F3-AVG	0.034243	0.053406	0.042737	1.000000	0.267592	0.279595
2	T3-AVG	0.035111	0.018115	0.037594	0.905651	0.215946	0.242483
3	T6-AVG	0.065352	0.025213	0.082383	0.031131	1.000000	0.240816
4	T4-AVG	0.047777	0.016862	0.061019	0.361978	0.547704	0.207068
5	F7-AVG	0.052317	0.007593	0.057103	0.339949	0.493534	0.190099
6	P4-AVG	0.009734	0.037142	0.029152	0.657678	0.207941	0.188329
7	F8-AVG	0.051927	0.078468	0.062365	0.154104	0.571532	0.183679
8	C4-AVG	0.005017	0.001454	0.027083	0.295319	0.421631	0.150101
9	F4-AVG	0.032134	0.018384	0.024173	0.395129	0.152699	0.124504

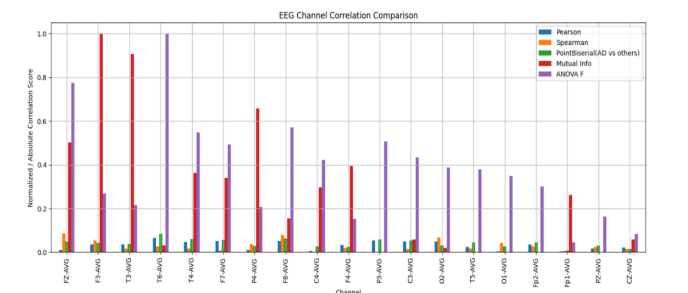


Figure 3: Results from testing out different correlations numerically and graphically

C. Model Architecture and Training

RestNET 1D Models are a specialised version of deep residual networks designed to work with one-dimensional sequential data, building on the core principles developed by He et al. for analysing signals that change over time. These networks use skip connections that allow information to flow directly through deep layers, solving a common problem where gradients become too small to effectively train very deep networks. The basic building block contains stacked one-dimensional convolution layers with normalisation and activation functions, where the processed signal $F(x)$ is added back to the original input x , creating the key identity shortcut $y = F(x) + x$. This structure enables the network to learn patterns at different time scales through stages that progressively reduce the signal length, with each block capturing local timing patterns while preserving information about the overall sequence. The one-dimensional convolutions typically use small filter sizes between 3 and 7 points, providing a good balance between processing speed and the ability to detect meaningful temporal patterns. ResNet-1D networks scale efficiently with longer sequences, making them well-suited for processing extended time-based signals in practical applications. The learned features can be understood through the layered structure, where initial layers detect simple timing details and deeper layers identify more complex temporal relationships. Performance can be further improved by incorporating techniques like dilated convolutions for wider time coverage, attention mechanisms for focusing on important parts, or multi-scale processing for handling different time resolutions. These advantages have made ResNet-1D a reliable choice for various applications, including time-series prediction, audio processing, and medical signal analysis, where understanding temporal patterns and maintaining computational efficiency are essential requirements.

2) *Custom RestNET Model*: The proposed experimental framework employs a 1D ResNet-18 backbone architecture, which is investigated for its ability to capture temporal dependencies in EEG signals through one-dimensional convolutional layers and residual connections. The residual connections are designed to preserve identity mapping across layers, potentially enabling stable gradient flow during training. Each residual block is configured to output a combination of the transformed input $F(x)$ and the original input x , mathematically expressed as:

$$\text{Output} = F(x) + x$$

To address the significant class imbalance present in the EEG dataset—where pathological classes such as MCI and AD are substantially underrepresented compared to healthy controls—a weighted training strategy is implemented as a necessary modification to the standard training procedure. Class weights are computed inversely proportional to class frequencies to examine whether rarer classes can achieve a larger impact on loss minimisation. The class weight for a class c is calculated as:

$$w_c = \frac{N}{C \cdot n_c}$$

where N represents the total number of samples, C denotes the number of classes, and n_c indicates the number of samples in class c .

During the optimisation process, the Adam optimiser is employed to investigate its effectiveness in adapting learning rates for each parameter using first and second moment estimates of gradients. The parameter update rule for each θ is given by:

$$\theta_{t+1} = \theta_t - \alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$$

This experimental design, combining residual-based architecture, class-aware training, and adaptive optimisation is hypothesised to enable the model to learn robust representations of EEG patterns, even under conditions of noise and imbalanced distributions. The effectiveness of these methodological choices will be evaluated through comprehensive performance metrics.

3) *Fine-tuned Pretrained Model*: The proposed framework utilises a 1D ResNet backbone designed to model the temporal dynamics of multichannel EEG signals. EEG data are represented as sequences with 22 input channels (corresponding to EEG, EKG, and auxiliary channels), and 1D convolutional layers are applied along the temporal axis. This architecture enables the model to extract temporal patterns while learning interactions across channels. The residual connections preserve input information and facilitate stable training by mitigating vanishing gradients. This design choice leverages the temporal dynamics inherent in EEG while preserving channel-specific information. The residual connections in the network implement identity mappings that facilitate stable gradient flow during training, mitigating the vanishing gradient problem commonly encountered in

deep networks. Each residual block outputs a combination of the transformed input $F(x)$ and the original input x , mathematically expressed as:

$$y=F(x)+x$$

where $F(x)$ represents the residual function learned by the convolutional layers within the block. This architecture enables the network to learn complex temporal features while maintaining efficient training dynamics.

Class Imbalance Handling:

Given the significant class imbalance in the EEG dataset—with pathological classes such as Mild Cognitive Impairment (MCI) and Alzheimer’s Disease (AD) underrepresented relative to healthy controls—a weighted loss function is employed. Class weights w_c are computed inversely proportional to class frequencies to amplify the contribution of rarer classes during loss minimisation:

$$w_c = \frac{N}{C \times n_c}$$

where N is the total number of samples, C is the number of classes, and n_c is the number of samples in class c . This weighting scheme encourages the model to learn discriminative features for minority classes, improving overall classification robustness.

Optimisation and Training Strategy:

The Adam optimiser is used for training, leveraging adaptive learning rates based on first and second moment estimates of gradients. The parameter update rule for each parameter θ is given by:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t$$

where m_t and v_t are bias-corrected estimates of the first and second moments of the gradients, η is the learning rate, and ϵ is a small constant for numerical stability.

III. RESULTS AND DISCUSSIONS

1) Custom Model : The 1D ResNet-18 model demonstrated moderate performance in classifying instances, achieving maximum validation accuracy of 53.67% and test accuracy of 54.95%. These reasonable scores indicate acceptable effectiveness in identifying instances, with macro average

recall of 0.53 and macro average precision of 0.53, suggesting balanced classification. When the model predicted an instance, it was correct around 53% of the time on average. These metrics showcase the model's ability to maintain equilibrium between identifying instances and minimising false positives. Overall, the 1D ResNet-18 model exhibits moderate performance, with balanced accuracy, recall, and precision across classes. The confusion matrix, shown in Figure 5, describes how well the 1D ResNet-18 model performed.

The model had reasonable accuracy predicting class 0, with balanced performance across metrics. However, there were classification challenges across all classes, with the model showing more consistent performance compared to previous results. For classes 1 and 2, the model classified instances with similar precision and recall rates (around 0.45-0.48), indicating consistent but moderate performance across pathological categories. For class 0, the model performed well with higher precision (0.69) and recall (0.67), showing better identification of normal instances while maintaining reasonable balance.

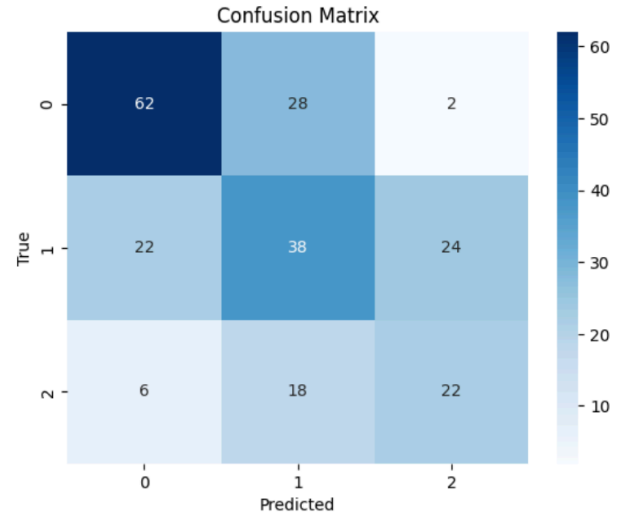


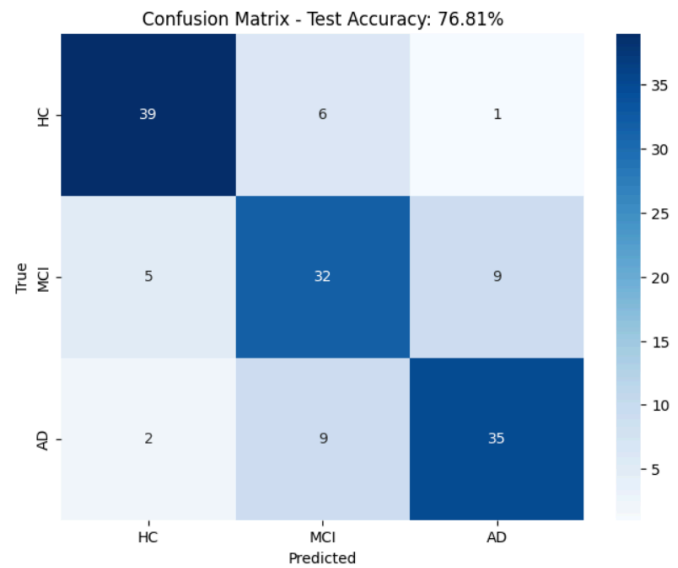
Fig. 5. Confusion matrix of custom model. Table 1

Classification Report:				
	precision	recall	f1-score	support
0	0.69	0.67	0.68	92
1	0.45	0.45	0.45	84
2	0.46	0.48	0.47	46
accuracy			0.55	222
macro avg	0.53	0.53	0.53	222
weighted avg	0.55	0.55	0.55	222

The classification report, shown in Table 1, provides comprehensive evaluation of the model's performance. Class 0 achieved good precision (0.69), recall (0.67), and F1-score (0.68), indicating reasonably accurate predictions for normal instances. Classes 1 and 2 showed more balanced patterns with similar precision and recall rates (class 1: 0.45 each; class 2: 0.46 precision, 0.48 recall), suggesting consistent though moderate identification of these diagnostic categories.

2) *Fine-Tuned Model* : The model achieved a final test accuracy of 76.81%, demonstrating strong performance in classifying EEG data into Healthy Controls, Mild Cognitive Impairment, and Alzheimer's Disease. As detailed in the classification report table (see Table 1), the precision and recall for Healthy Controls were both 0.85, while for Alzheimer's Disease, precision reached 0.78 and recall 0.76. The Mild Cognitive Impairment class, which is clinically more ambiguous, showed a precision of 0.68 and recall of 0.70, with a macro-averaged F1-score across all classes of 0.77. Readers are encouraged to consult Table 1 for a comprehensive breakdown of these metrics. Additionally, the confusion matrix (see Figure 1) visually illustrates the distribution of correct and incorrect predictions, highlighting the model's high accuracy in identifying healthy individuals and the specific patterns of misclassification between MCI and AD. Figure 1 provides a clear, visual summary of the model's decision boundaries and reinforces the numerical findings presented in Table 1, offering a holistic view of the model's classification capabilities¹².

Classification Report:				
	precision	recall	f1-score	support
HC	0.85	0.85	0.85	46
MCI	0.68	0.70	0.69	46
AD	0.78	0.76	0.77	46
accuracy			0.77	138
macro avg	0.77	0.77	0.77	138
weighted avg	0.77	0.77	0.77	138



IV. CONCLUSIONS

Effective and accurate diagnosis of AD is important to initiate effective treatment and early investigation of AD is expected to play an immense role in the development of cures. This study explored the effectiveness of deep learning models, specifically a custom 1D ResNet-18 and a fine-tuned pretrained ResNet-18, for classifying Alzheimer's disease stages using EEG signals. The custom 1D ResNet-18 model demonstrated moderate performance, achieving balanced but limited accuracy, precision, and recall across healthy controls and pathological classes. Contrastingly, the fine-tuned pretrained ResNet-18 model significantly outperformed the custom model, achieving a test accuracy of 76.81% along with improved precision and recall metrics, particularly for healthy controls and Alzheimer's disease classes. In conclusion, this study demonstrates that EEG-based neurodegenerative disease classification can be significantly enhanced through residual architectures and multi-stage training, with patient age as auxiliary input notably improving model performance and underscoring the clinical value of demographic information.

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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